Robust and Uncertainty-Aware VAE (RU-VAE) for One-Class Classification

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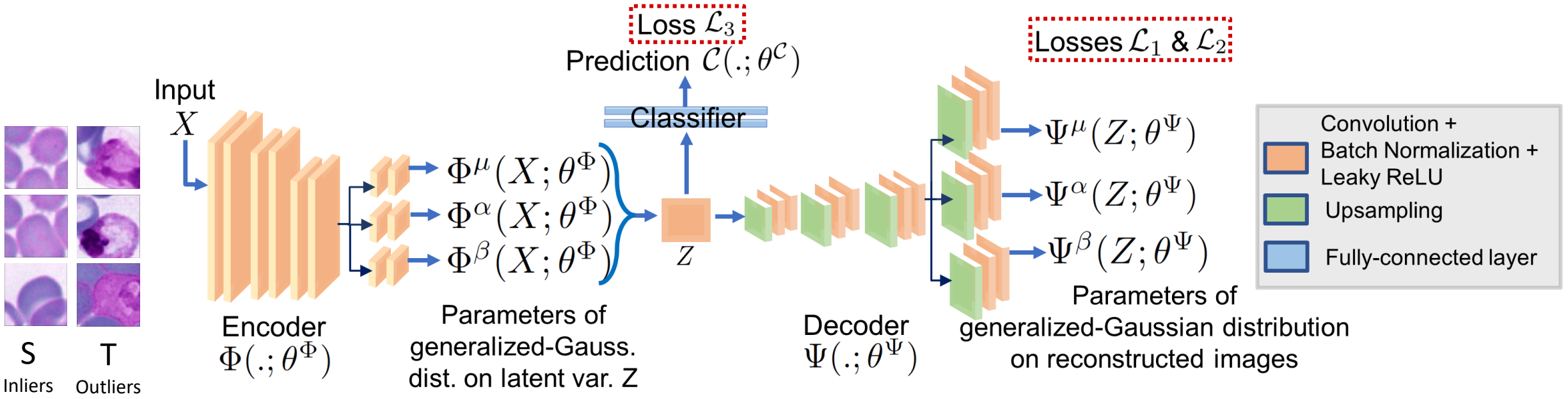
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Introduction

- Leveraging generalized Gaussian (GG) models on both
  - encodings in the latent space and
  - the reconstructed image
- Data adaptive modeling subsuming
  - robust modeling (through the GG's shape parameters) and
  - Uncertainty-aware modeling (through the GG's scale parameters)
- Reparameterization of the GG
- Classifier in the latent space

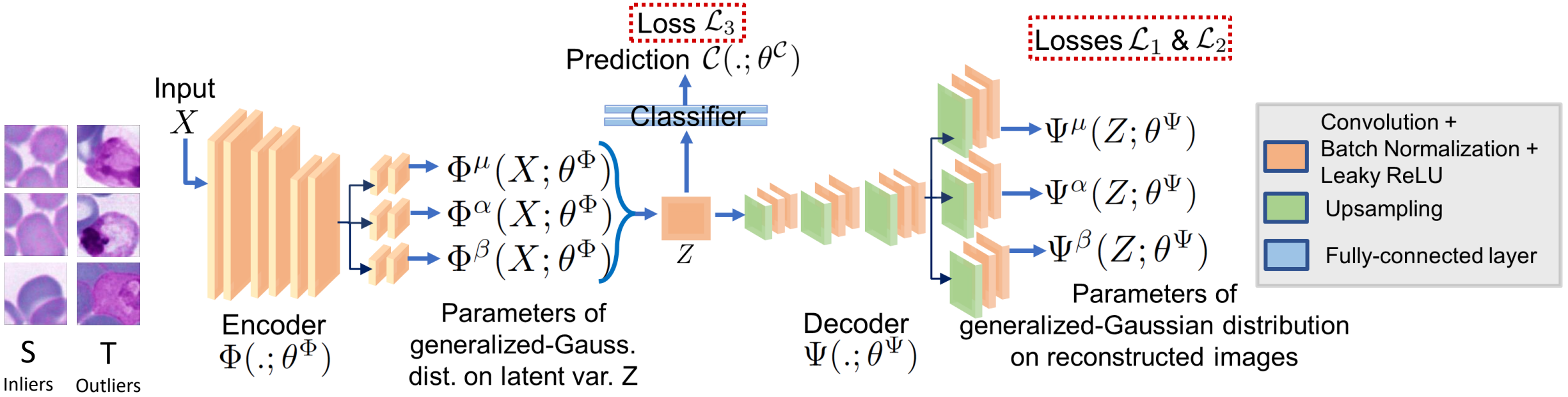


RU-VAE Framework



$\boldsymbol{\mu}$: Mean parameter of generalized-Gaussian distribution
$\boldsymbol{\alpha}$: Scale parameter of generalized-Gaussian distribution
$\boldsymbol{\beta}$: Shape parameter of generalized-Gaussian distribution

Objective Function



Inliers

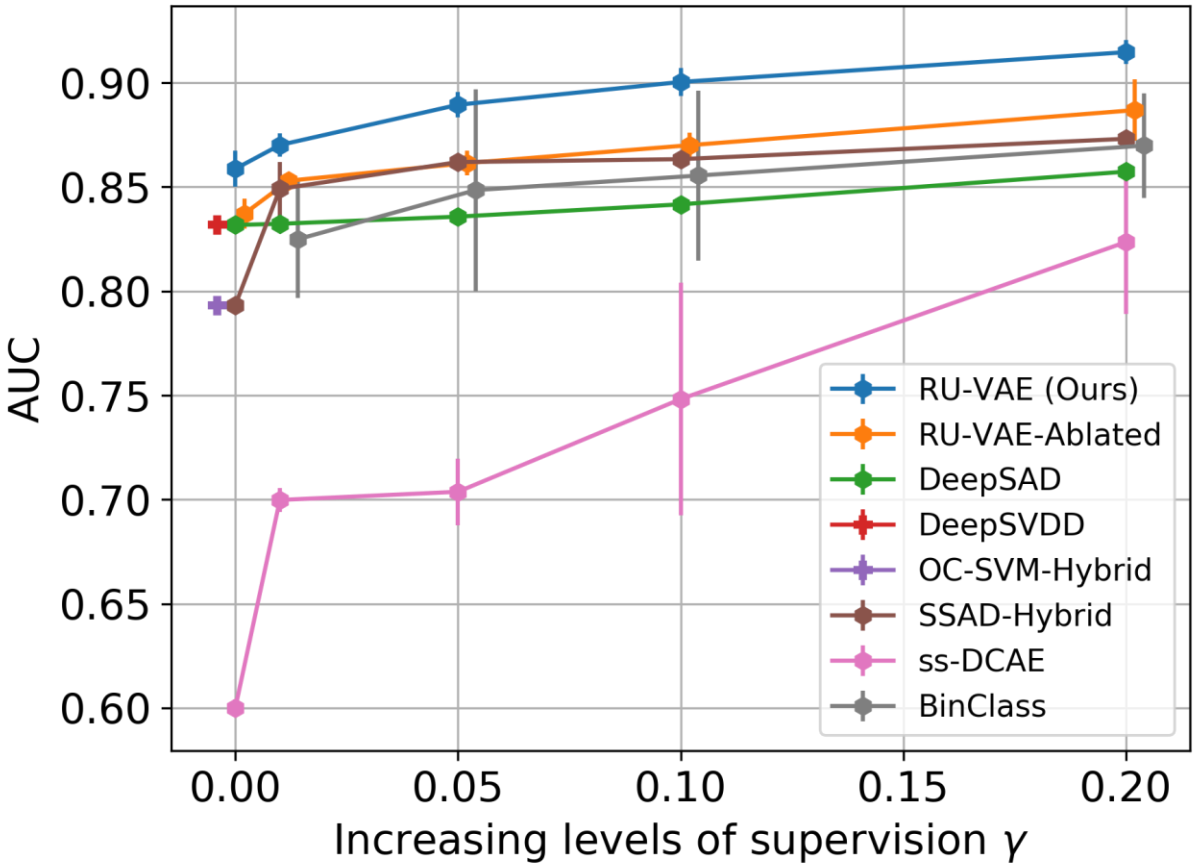
Outliers

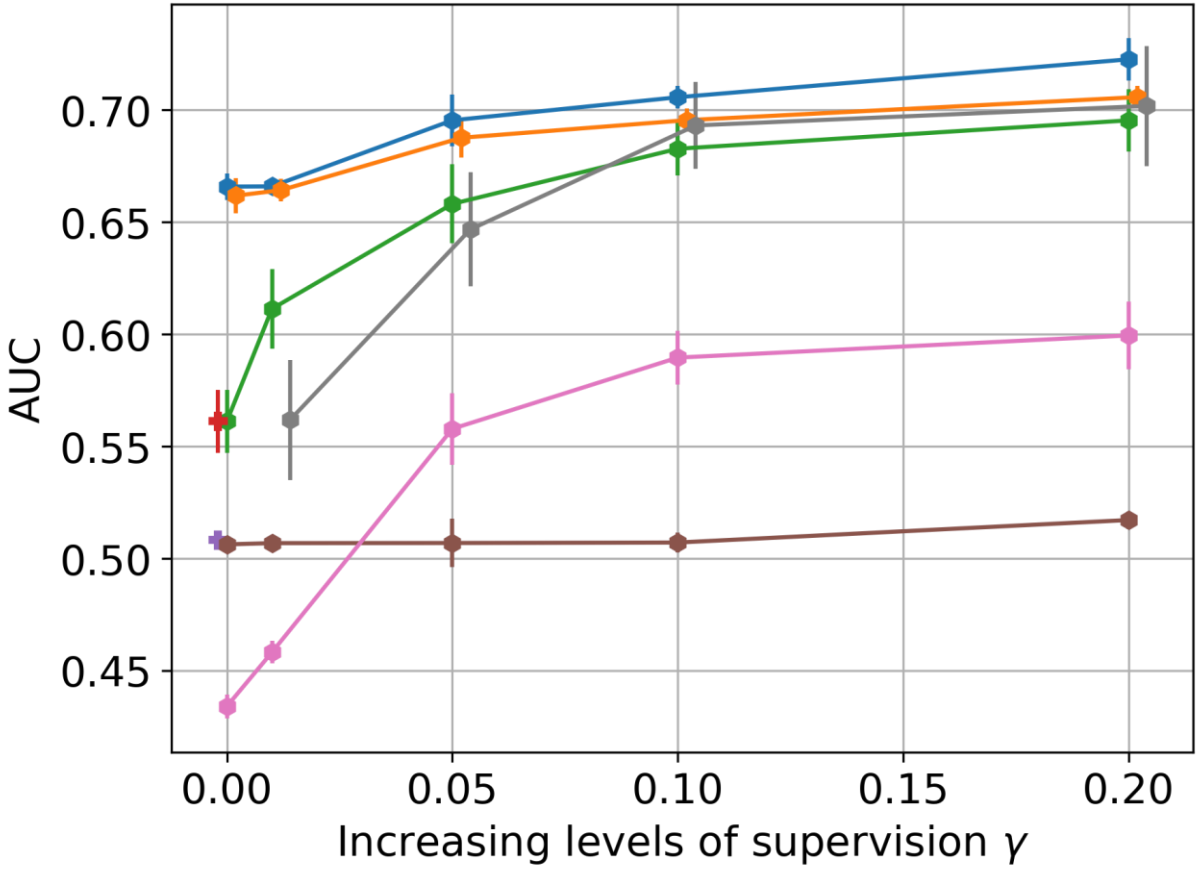
$$\lambda_1 \mathcal{L}_1(\{S_n\}_{n=1}^{N};\theta^{\Phi\Psi}) + \lambda_2 \mathcal{L}_2(\{T_m\}_{m=1}^{M};\theta^{\Phi\Psi}) +$$

$$\mathcal{L}_3(\{S_n\}_{n=1}^{N},\{T_m\}_{m=1}^{M};\theta^{\Phi\Psi}),$$

Classifier

Results on Malaria and Nanofibre datasets





Results on Malaria datasets

